

Airline Sentiment Insights Report

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Summary

The tool rated 49.0% of 14,604 airline tweets as negative, against about 63% in the dataset's own labels, so it still understates complaints and is best used to compare airlines and days, not to state absolute levels.

- **Overall:** 32.7% positive, 18.3% neutral, 49.0% negative.
- **Worst airline:** US Airways, with 60.9% negative tweets. Virgin America (33.3%, but only 504 tweets) and Delta (36.2%) are lowest.
- **Main complaints:** customer service (31.2% of recorded reasons), then late flights (21.0%) and cancelled flights (11.8%).
- **Trend:** negative sentiment rose from 40.3% on 18 Feb to a peak of 57.3% on 22 Feb 2015, then eased to 47.6% by 24 Feb.
- **Reliability:** the first version of the tool agreed with the dataset's labels on 55.0% of tweets. After adding a customer-complaint word list it agrees on 65.9%, and it finds 69% of the tweets labelled negative, up from 51%.

About the data and method

The data is the public Kaggle "Twitter US Airline Sentiment" file: 14,640 tweets about six US airlines, posted 16 to 24 February 2015. It has 15 columns, including the tweet text, the airline, the date and a reason for each negative tweet.

The tool checked and cleaned the file before scoring. It removed 36 exact duplicate rows (14,640 down to 14,604), tidied extra spaces in 2,339 cells and read all 14,604 dates. It found no blank texts and no spelling variants to merge.

Each tweet was scored with VADER, a rule-based method that rates words from -1 (very negative) to +1 (very positive). A score of 0.05 or more is positive, -0.05 or less is negative, and anything between is neutral. Links, @mentions and hashtag symbols are removed first, and web codes left over from tweets (such as the one that stands for the & sign) are turned into real characters.

The first run used plain VADER. A second run used the tool's customer messages mode, which adds a hand-written list of common complaint words (such as cancelled, delayed, lost, hold) and treats "help" and "please" as neutral. All results below use the second run, and the reliability section compares the two.

Overall sentiment

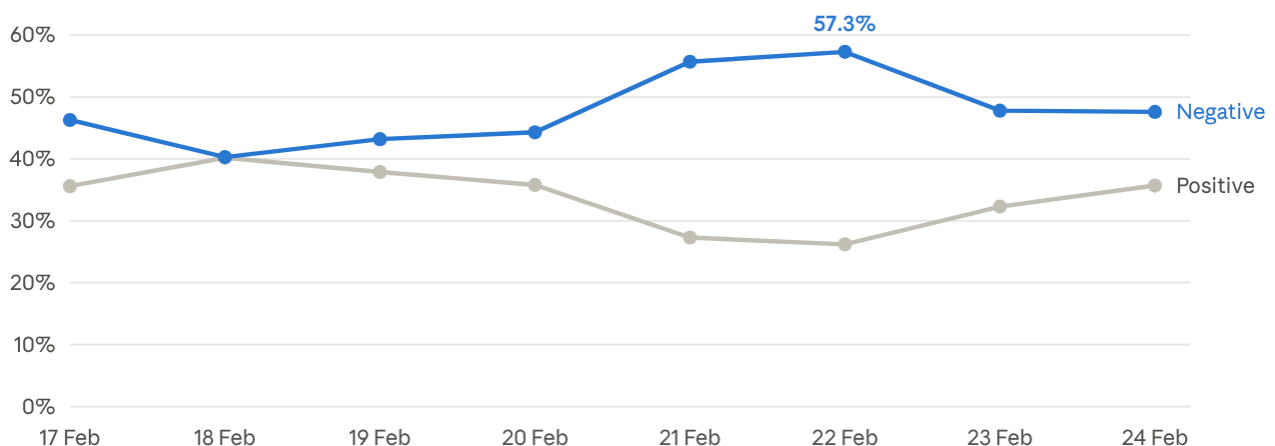
Negative is now the largest group, at about half of all tweets.

Sentiment	Tweets	Share
Positive	4,778	32.7%
Neutral	2,669	18.3%
Negative	7,157	49.0%

Sentiment over time

Negative tweets peaked at 57.3% on 22 Feb, when volume doubled

Share of each day's tweets, 17 to 24 February 2015



Tool results from the Kaggle Twitter US Airline Sentiment file, 17 to 24 Feb 2015. 16 Feb left out (4 tweets).

The share of negative tweets climbed from 40.3% on 18 Feb to 57.3% on 22 Feb. Tweet volume doubled at the same time, from about 1,400 a day to over 3,000 on 22 and 23 Feb. Negative tweets outnumbered positive ones on every day in this run, and across the whole period the negative share rose from 43.3% in the first half to 51.3% in the second.

Sentiment by airline

US Airways has the highest share of negative tweets (60.9%) and Virgin America the lowest (33.3%), but Virgin America has only 504 tweets, so its figure is less certain.

Airline	Tweets	Positive	Neutral	Negative
US Airways	2,913	23.9%	15.3%	60.9%

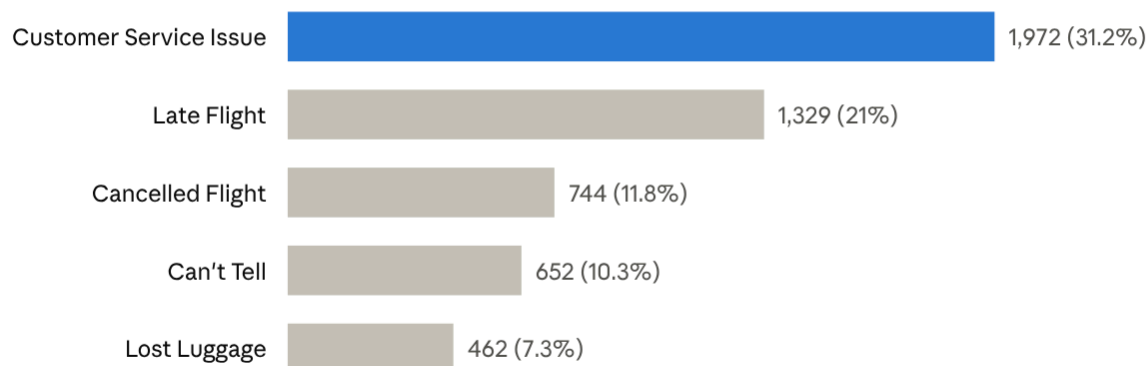
Airline	Tweets	Positive	Neutral	Negative
American	2,723	29.3%	16.7%	53.9%
United	3,822	31.7%	17.6%	50.7%
Southwest	2,420	39.6%	18.8%	41.6%
Delta	2,222	40.5%	23.3%	36.2%
Virgin America	504	42.5%	24.2%	33.3%

The ranking held up between the two runs. US Airways, American and United were the three highest both times, and Southwest, Delta and Virgin America were the three lowest, though their order changed, so treat those three as roughly level. United has the most tweets, so it carries the most complaints in absolute terms.

Why customers are unhappy

Customer service is the top reason for negative tweets, ahead of late flights

Tweets by recorded reason: the five largest. Percent is the share of all recorded reasons.



Tool results from the same file. Reasons are recorded only for negative tweets; the three smallest reasons are left out.

Late flights (1,329) and cancelled flights (744) together make up 32.8% of recorded reasons, slightly more than customer service alone (31.2%). "Can't Tell" (10.3%) means the dataset could not name a clear reason.

How reliable is the tool

The first version of the tool agreed with the dataset's labels on 55.0% of the 14,604 tweets, and tuning it for customer messages raised that to 65.9%. The first version was too positive: the labels mark about 63% of tweets as negative, and the tool found only

about half of them.

Version	Agrees with labels	Negative tweets found	Tweets rated negative
Plain VADER	55.0%	51%	35.5%
Customer messages mode	65.9%	69%	49.0%

The table below shows the final version against the dataset's labels.

Label in the dataset	Tweets	Tool says positive	Tool says neutral	Tool says negative
Positive	2,354	2,021	179	154
Neutral	3,091	1,118	1,286	687
Negative	9,159	1,639	1,204	6,316

It finds 86% of positive tweets, 69% of negative ones and 42% of neutral ones. The remaining errors are of two kinds:

- **Polite complaints.** "Thanks" and "thank" are still the most common words in the tweets the tool rated positive, and customers also use them in complaints. 1,639 negative tweets are still scored positive.
- **Neutral tweets.** 1,118 tweets labelled neutral, mostly questions and statements, are scored positive, and 687 are scored negative.

Two cautions apply. The complaint word list was written after looking at the most common words in this file, so the 65.9% is measured on data that informed the list, and a smaller gain is likely on other data. And the airline ranking is still the safer result than the absolute percentages, because the same bias applies to every airline.

Recommendations

For an airline reading these results:

1. **Fix customer service first.** It is the largest recorded reason for negative tweets (31.2%), ahead of any single operational cause.
2. **Treat delays and cancellations as one problem.** Late and cancelled flights together make up 32.8% of recorded reasons.
3. **Look hardest at US Airways.** It has the highest share of negative tweets, 60.9%, and 2,913 tweets is enough to trust the gap.

4. **Watch for daily spikes.** The 22 Feb peak came with a doubling of tweet volume, so a rising count of tweets is an early warning.

For improving the tool:

1. Test a stronger model, such as one trained on tweets, and compare its agreement with the 65.9% here.
2. Check the tool on a second labelled dataset, to see how much of the gain holds up outside airline tweets.
3. Handle sarcasm and emoji, which a word-based method cannot read.

Limitations and next steps

- The data covers only nine days in February 2015, and the first day has just 4 tweets, so the trend is short and the first point is not meaningful.
- The data cannot say why negative sentiment peaked on 22 Feb. Weather or an outage would be a guess; an airline would need to check its own operations records.
- The reason column is filled in only for tweets the dataset labels negative, so the reasons describe the 6,316 tweets that both the tool and the dataset call negative, not every complaint.
- Even after tuning, the tool rates fewer tweets negative (49.0%) than the dataset's labels do (about 63%), so absolute shares are understated.
- The tool works on English words and misses sarcasm and context, as the accuracy check shows.